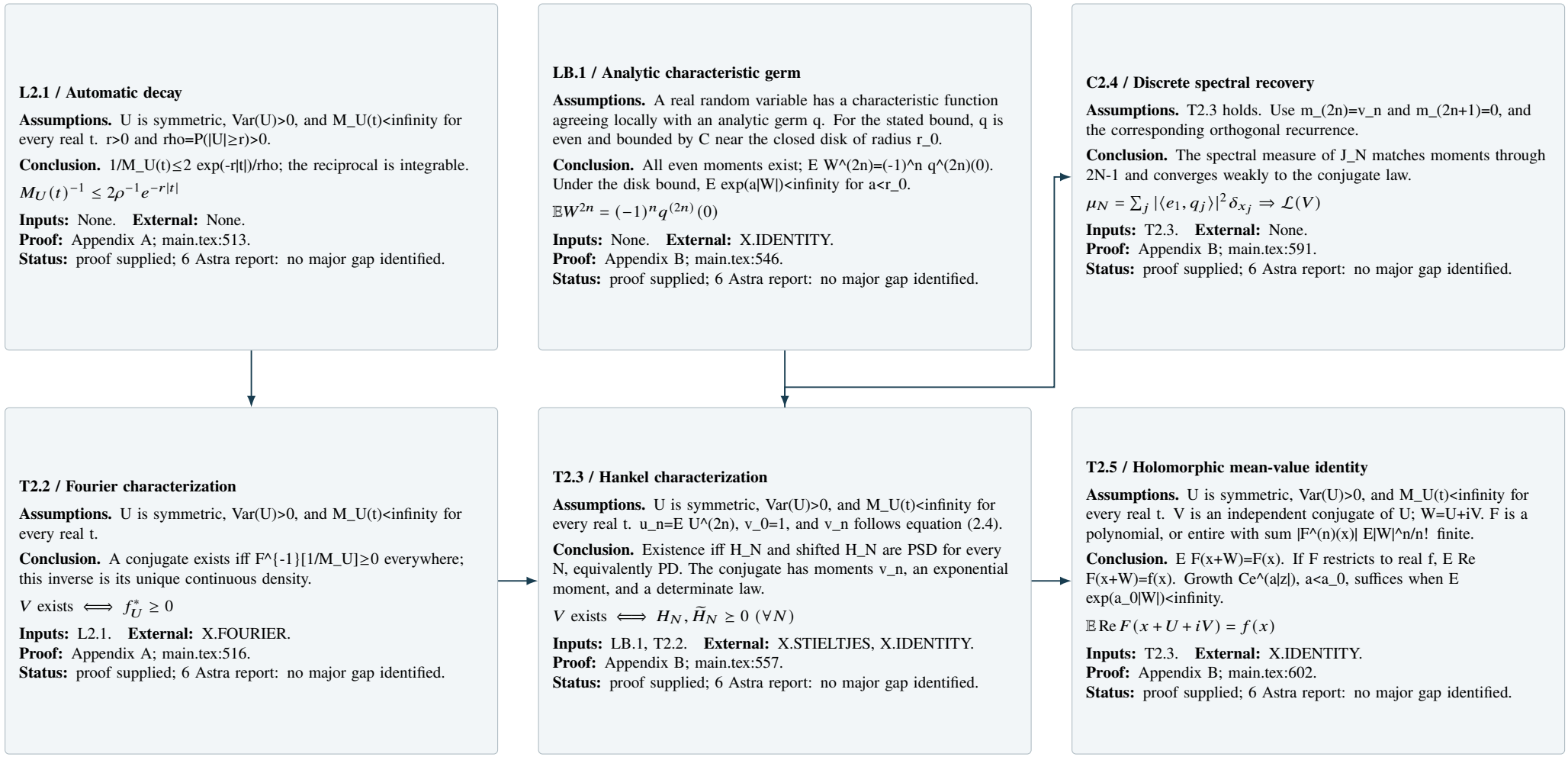
Theorem graph / 1. Existence, recovery, and the mean-value identity

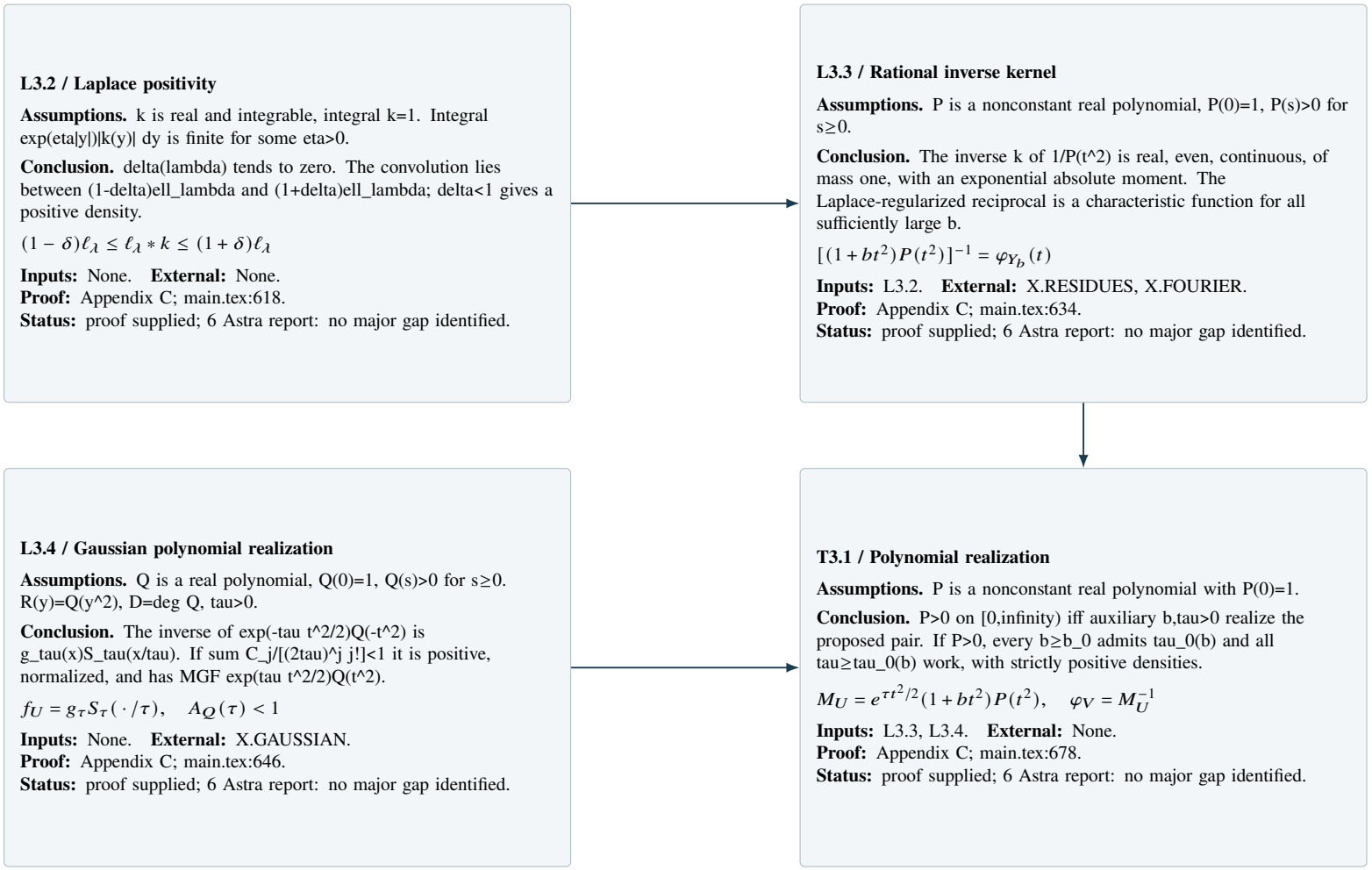
Arrows show dependencies within this panel. Other inputs are listed by ID on each card. Full records: `theorem_graph/theorems.json`.



External dependency status. X.FOURIER: classical; applied and detailed in Appendix A. X.IDENTITY: classical; applied with explicit domains. X.STIELTJES: classical; cited, not reproved.

Theorem graph / 2. Polynomial realization

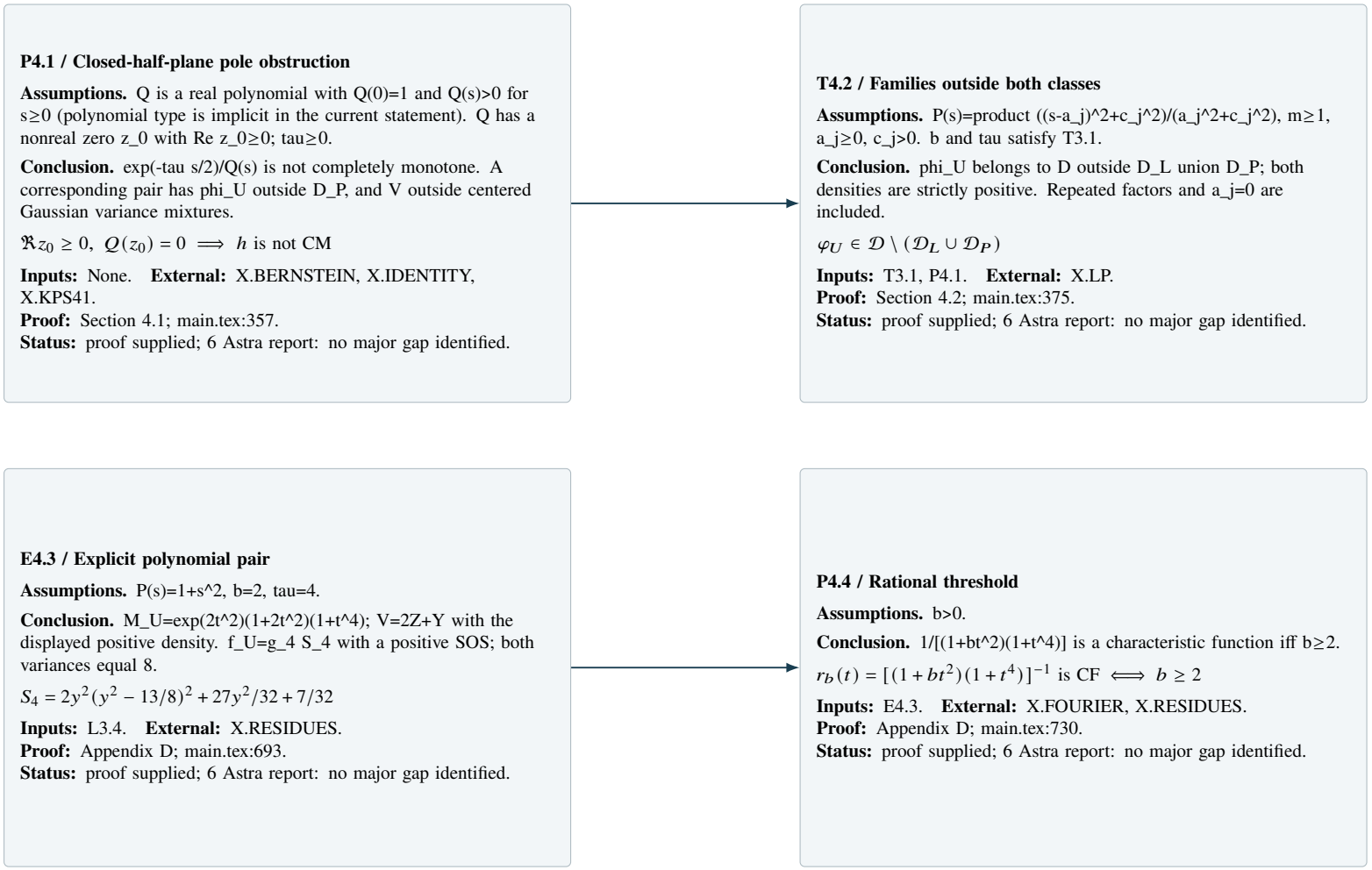
Arrows show dependencies within this panel. Other inputs are listed by ID on each card. Full records: `theorem_graph/theorems.json`.



External dependency status. X.FOURIER: classical; applied and detailed in Appendix A. X.GAUSSIAN: classical; derivative formula derived in Appendix C. X.RESIDUES: classical; contour argument given in Appendix C.

Theorem graph / 3. Structural exclusions and the explicit family

Arrows show dependencies within this panel. Other inputs are listed by ID on each card. Full records: `theorem_graph/theorems.json`.



External dependency status. X.BERNSTEIN: classical; cited, not reproved. X.FOURIER: classical; applied and detailed in Appendix A. X.IDENTITY: classical; applied with explicit domains. X.KPS41: primary statement checked; used as an external theorem. X.LP: classical representation used to identify the comparison class. X.RESIDUES: classical; contour argument given in Appendix C.

Theorem graph / 4. Recovery of known classes

Arrows show dependencies within this panel. Other inputs are listed by ID on each card. Full records: `theorem_graph/theorems.json`.

P5.1 / Independent-sum recovery

Assumptions. U is symmetric with M_U(t)=exp(sigma^2 t^2/2) product(1+t^2/a_n^2). sigma≥0, a_n>0, sum a_n^(-2)<infinity; the product is a valid MGF.

Conclusion. The independent sum V=G+sum L_n converges in L2 and almost surely and is a conjugate. For nondegenerate U its density is the Fourier inverse.

$V = G + \sum_n L_n, \quad L_n \sim \mathrm{Laplace}(0, 1/a_n)$

Inputs: T2.2. **External:** X.L2, X.LP.
Proof: Appendix F; main.tex:767.
Status: proof supplied; 6 Astra report: no major gap identified.

P5.2 / Bernstein mixture recovery

Assumptions. Psi belongs to N_D as specified in Section 5.2. Use the characteristic-function and Bernstein assertions of KPS Theorem 4.1.

Conclusion. There is T_Psi>0 with E exp(-sT)=exp(-Phi_Psi(s)); V=sqrt(2T)Z is the conjugate, with finite continuous Gaussian-mixture density and E T^(-1/2)<infinity.

$V_\Psi = \sqrt{2T_\Psi}Z, \quad f_V(x) = \mathbb{E}\frac{e^{-x^2/(4T_\Psi)}}{\sqrt{4\pi T_\Psi}}$

Inputs: LB.1, L2.1, T2.2. **External:** X.KPS41, X.BERNSTEIN.
Proof: Appendix F; main.tex:785.
Status: proof supplied; 6 Astra report: no major gap identified.

External dependency status. X.BERNSTEIN: classical; cited, not reproved. X.KPS41: primary statement checked; used as an external theorem. X.L2: classical; applied in Appendix F. X.LP: classical representation used to identify the comparison class.

Reading the registry. Every result has a stable ID, hypotheses, conclusion, proof and statement locations, internal dependencies, external dependency status, and contribution role. Source hashes bind the graph to the manuscript version.

X.KPS41 supplies the Bernstein representation for $\mathcal{D}_P$. X.LP supplies the classical product representation for class identification; P5.1 assumes that the displayed product is a valid moment-generating function.

Review status. The user-supplied 6 Astra report identifies no major mathematical gap. The two recorded editorial clarifications and the outstanding historical comparison are retained in the JSON.